Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials and Rational Extension of Various Quantum Mechanical Systems

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ABSTRACT

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Abstract

In quantum mechanics (QM), exactly solvable (ES) potentials hold a significant place because they give the explicit solution to the corresponding Schrödinger equation, which provides all information and deep understanding of the associated system. The explicit solutions derived from ES potentials may be used to compute a range of physical parameters, including transition probabilities, expectation values, and scattering cross sections. One of the most notable advancements in quantum physics in recent years has been the discovery of new ES systems, which play a crucial role in both theoretical and experimental aspects. The ES potentials work as standard models to help in evaluating the precision and validity of different approximation techniques. ES potentials provide the extraordinary ability to approximate complex systems that cannot be solved exactly. The exact solutions for quantum mechanical systems provide a helpful beginning point for perturbation theory or variational approaches. However, the number of ES potentials documented in the literature remains limited, with examples including the Coulomb potential, simple harmonic oscillator potential, Morse potential, Scarf II, Pöschl-Teller, and Eckart potentials, among others.[1] Due to their significance, researchers have devoted significant effort to developing new ES potentials. For this, approaches include supersymmetric (SUSY QM), shape-invariant potentials (SIPs), and algebraic methods such as the factorization method. These methods often build on classical techniqus or introduce novel mathematical tools to discover and classify new ES potentials. For example, SUSY QM offers a framework for constructing new ES potentials by connecting them to known ones through SUSY transformations. There has been a resurgence of interest in finding exact solutions in QM, driven by the discovery of a new type of spectral problem called quasi-exactly solvable (QES) problems.[2] The unique aspect of these problems is that a finite number of initial eigenstates can be explicitly determined using algebraic methods. In other words, only the initial portion of the spectrum can be obtained, rather than the entire spectrum. The remaining special states are complex and can only be approximated. This discovery has refined the definition of an exact solution and established a framework for identifying problems that admit such solutions. In ES systems, solutions are typically expressed in terms of hypergeometric or confluent hypergeometric functions, which can be further reduced to well-known classical orthogonal polynomials such as Jacobi, Laguerre, and Hermite. The significance and wide-ranging application of these COPs have been thoroughly studied and are prevalent in many areas of physics.

[1]F. Cooper, Avinash khare and Uday Sukhatme, Supersymmetry in Quantum Mechanics, World Scientific, (2001).

[2]Alexander G Ushveridze, Quasi-exactly solvable models in quantum mechanics, (2017).

In 2009, a significant advancement in the field of orthogonal polynomials was made with the introduction of two new families, the X_1-Jacobi and X_1-Laguerre exceptional orthogonal polynomials (EOPs) [3]. These EOPs, which are related to COPs, satisfy a Sturm-Liouville problem. Unlike classical orthogonal polynomials, the sequence of these EOPs begins with a degree of $n = 1$ rather than a constant. Nevertheless, they still form a complete set of orthonormal basis functions with respect to a positive definite measure. This discovery spurred a series of developments, leading to the generalization of the X_1 case into the X_m-EOPs and their multi-index extensions. The sequence of X_m-EOPs begins with a degree of m, where m is a positive integer that can be $1, 2, 3, \ldots$.

Following the discovery of these EOPs, researchers expanded the list of ES potentials and derived their bound state wavefunctions using these EOPs [4]. These newly discovered potentials, whose bound state solutions are expressed in terms of EOPs, are referred to as rationally extented potentials. The list of ES potentials includes rationally extended generalized Pöschl-Teller, trigonometric Scarf, and radial oscillator potentials. A particularly intriguing feature of these rationally extended potentials is that although their eigenfunctions differ completely from those of traditional potentials, they are strictly isospectral. This raises an interesting question about the key reasons behind their isospectrality.

In QM, the concept of position-dependent mass (PDM) has intrigued researchers over the past thirty years. Unlike traditional quantum systems where the particle mass is constant, PDM systems feature particles with mass that varies based on their position. This unique characteristic has numerous applications, making it a significant area of study. The exploration of PDM systems began with O. von Roos, who introduced a generalized form of the kinetic energy operator [5]. Recent advancements in the study of PDM systems have utilized the generalized von Roos Hamiltonian and point canonical transformations, leading to exact solutions for certain PDM systems. A notable contribution comes from R. N. Costa Filho and his colleagues [6], who introduced a new approach to tackling PDM problems by employing a generalized translation operator. This method facilitates small nonlinear displacements, offering fresh insights into PDM systems. These advancements have been particularly beneficial in condensed matter physics, enhancing our understanding of quantum liquids, quantum dots, and other semiconductor structures. Additionally, PDM systems play a crucial role in the study of nuclei, impurities in crystals, metal clusters, quantum liquids, and semiconductor heterostructures.

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[3] D. Gomez-Ullate, N. Kamran, and R. Milson, J. Math. Anal. Appl. 359 (2009) 352.

[4] C. Quesne, Journal of Physics A: Mathematical and Theoretical 41.39 (2008), p. 392001.

[5] O. von Roos, Physical Review B 27 (12) (1983) 7547.

[6] R. Costa Filho, M. Almeida, G. Farias, J. Andrade Jr, Physical Review A-Atomic, Molecular, and Optical Physics 84 (5) (2011) 050102

Following the discovery of EOPs, Midya et al.[7] derived exactly solvable potentials for the PDM Schrödinger equation. They demonstrated that the wavefunctions corresponding to these exactly solvable potentials can be expressed in terms of X_1-Laguerre or X_1-Jacobi EOPs. These potentials represent generalizations, through certain rational functions, of previously obtained potentials[8], whose bound state solutions involve classical Laguerre or Jacobi orthogonal polynomials. By examining this scenario within the framework of supersymmetry (SUSY), they showed that these potentials are SIPs and isospectral to the previously obtained potentials in a PDM setting, where the solutions are described by classical Laguerre or Jacobi orthogonal polynomials. This finding opens the door to extending the analysis to find various exactly solvable potentials in the PDM context, whose wavefunctions are formulated in terms of X_m-EOPs, and to discover new SIPs in different types of PDMs for which the solutions can be expressed using X_m-EOPs.

There has been growing interest in finding exact solutions at zero energy[9] $(E = 0)$, driven by advances in SUSY QM[10] and the quest for conditionally and quasi-exactly solvable problems. This makes it important to identify zero modes in quantum mechanical systems. Zero-energy solutions are particularly intriguing because they can only be solved at this energy level. Contrary to common belief, these solutions can be as dynamic and significant as other energy states, showing notable kinetic energy at finite distances from the interaction center.

In the study of quantum systems with non-confining potentials, the emphasis is often on bound states $(E < 0)$ and free states $(E > 0)$. However, zero energy states $(E = 0)$ possess unique properties that are frequently overlooked. In some systems, such as the Coulomb problem, the $E = 0$ state is non-normalizable and part of the continuum. Nevertheless, there are notable exceptions where the $E = 0$ state is bound and normalizable for specific values of the coupling constant. These findings have significant theoretical and practical implications, challenging the assumption that all zero-energy states reside in the continuum. The existence of bound $E = 0$ states broadens our understanding of quantum systems and opens new research avenues. Thus, it is intriguing to present another example of a zero-energy quantum state corresponding to some QES potentials.

Details of these works are given below.

Chapter 1 offers a concise introduction to ES and QES systems, highlighting the significant advancements in this area of research. It explores key concepts such as SUSY, shape invariance (SI), and the factorization method, which have been instru-

[7] B. Midya, B. Roy, Phys. Lett. A 373 (45) (2009) 4117.
[8] B. Bagchi, P. Gorain, C. Quesne, R. Roychoudhury, Europhys. Lett., 72 (2005), p. 155
[9] J. Daboul, M. M. Nieto, Physics Letters A 190 (5-6) (1994) 357–362
[10] B. Bagchi, C. Quesne, Physics Letters A 230 (1-2) (1997) 1–6.

mental in enhancing the understanding of these systems. Additionally, we delve into the importance and historical context of PDMs and EOPs. Following this foundation, the motivations driving our research are discussed, along with a concise summary of the work conducted in this thesis.

In **chapter 2**, we introduce essential mathematical tools that are foundational for the rest of the thesis. We begin with EOPs, discussing their differential equations, specific examples of X_1-Jacobi and X_1-Laguerre polynomials, their corresponding weight functions, and some lower degree polynomials. We then generalize these differential equations to more general cases, namely X_m-Jacobi and X_m-Laguerre polynomials. We also provide a brief overview of the techniques used in SUSY QM, including the construction of a hierarchy of Hamiltonians and SIPs. Following this, we briefly discuss mathematical tools such as the PCT approach and the group theoretic approach.

In **chapter 3**, we address the fundamental question of why rationally extended potentials are strictly isospectral to the usual potentials. We answer this by considering the X_1-Jacobi and X_1-Laguerre differential equations as eigenvalue equations to define Hamiltonians and analyze them within the SUSY framework[11]. By constructing a hierarchy of Hamiltonians that exhibit SI, we determine the eigenvalues of the original Hamiltonian. We demonstrate that the solubility of these differential equations arises from the underlying SI symmetry. We then extend our study to X_m-Jacobi and X_m-Laguerre polynomials, showing that the solubility of these differential equations is due to both SUSY and SI symmetries. These symmetries are the reasons for the isospectrality of rationally extended potentials, where the wavefunctions are expressed in terms of these EOPs.

In **chapter 4**, we begin with the PCT approach in a PDM background to construct ES potentials. We solve the Schrödinger equation with two different PDMs. In the first case, we identify the ES potential whose eigenfunctions are expressed in terms of X_m-Laguerre polynomials. In the second case, we introduce a new PDM that depends on a parameter ν [12]. This leads to a one-parameter family of potentials, with the corresponding eigenfunctions also expressed in terms of X_m-Laguerre EOPs. Subsequently, we employ SUSY techniques to find the SIPs corresponding to the potentials in both cases. We find that the exact solvability of these potentials is due to SI symmetry.

In **chapter 5**, we focus on finding normalizable zero-energy quantum states [13]. We investigate a rationally extended truncated Calogero-Sutherland model by applying the PCT approach to the associated Schrödinger equation. By matching the resulting energy expression with the Casimir of the $SO(2,1)$ algebra, we identify three classes of QES potentials. For each class, with appropriate restrictions, we find normalizable

[11]Satish Yadav, Avinash Khare, and B. P. Mandal, Annals of Physics 444, 169064 (2022)

[12]Satish Yadav, Rahul Ghosh, and B. P. Mandal, arXiv preprint arXiv:2401.00995 (2024)

[13]Satish Yadav, Sudhanshu Shekhar, Bijan Bagchi, and B. P. Mandal, arXiv preprint arXiv:2406.09164 (2024)

wavefunctions that are asymptotically smooth.

In **chapter 6**, we summarize the results presented in the various chapters of this thesis and discuss future prospects for our research.

List of Publications

1. **Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials**
   Satish Yadav, Avinash Khare and B. P. Mandal
   Annals of Physics 444, 169064 (2022)
   DOI: https://doi.org/10.1016/j.aop.2022.169064

2. **New solutions of Isochronous potentials in terms of exceptional orthogonal polynomials in heterostructures**
   Satish Yadav, Rahul Ghosh and B. P. Mandal
   arXiv preprint arXiv:2401.00995(2024)
   DOI: https://doi.org/10.48550/arXiv.2401.00995

3. **Generating QES potentials supporting zero energy normalizable states for an extended class of truncated Calogero Sutherland model**
   Satish Yadav, Sudhanshu Shekhar, Bijan Bagchi and B. P. Mandal
   arXiv preprint arXiv:2406.09164(2024)
   DOI: https://doi.org/10.48550/arXiv.2406.09164

4. **Supersymmetry and Shape Invariance of Exceptional Hermite Polynomials.**
   Satish Yadav, Avinash Khare and B. P. Mandal
   To be submitted

5. **Master Function Associated with Exceptional Orthogonal Polynomials**
   Satish Yadav, Avinash Khare and B. P. Mandal
   In Prepration